

THERMODYNAMIC PROPERTIES OF THE GADOLINIUM GALLIUM GARNET, $\text{Gd}_3\text{Ga}_5\text{O}_{12}$, BETWEEN 0.05 AND 25 K

B. DAUDIN, R. LAGNIER and B. SALCE

Centre d'Etudes Nucléaires de Grenoble, Service des Basses Températures, Laboratoire de Cryophysique, 85 X, 38041 Grenoble Cedex, France

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The specific heat, the entropy, the thermal conductivity and the diffusivity of $\text{Gd}_3\text{Ga}_5\text{O}_{12}$ have been measured in the presence of an applied magnetic field. Calculations have been performed in order to explain simultaneously all these results within the framework of a simple molecular field theory. The agreement is good above 5 K and at low temperatures the deviations may be explained by the limitations of the simple model.

1. Introduction

The development of industrial low temperature devices has led to an increase in interest in continuous magnetic cooling, and in particular in machines with a high cooling power. Such a machine with a high efficiency and a high cooling power was recently built in Grenoble and works with gadolinium gallium garnet (GGG) as the active magnetic element [1,2].

In this paper we present measurements of thermal conductivity, specific heat and entropy of this compound as a function of temperature and magnetic field. All experiments were carried out with samples of the same origin as those used by Lacaze et al. [2].

The thermal conductivity [3] (see fig. 1), specific heat [4–7] and magnetothermodynamic properties [5–7] of this compound were previously measured some ten years ago by various workers.

The new features of our work compared to these previous ones are as follows:

- i) the temperature range of thermal conductivity measurements was extended down to a lower temperature (50 mK) and magnetic field was applied,
- ii) the temperature range of specific heat measurements was extended,

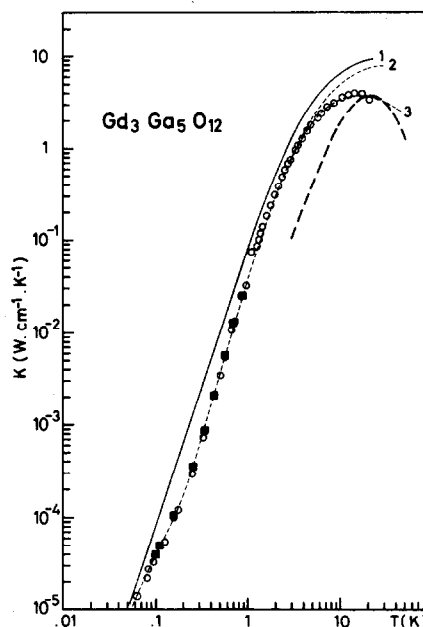


Fig. 1. Thermal conductivity of $\text{Gd}_3\text{Ga}_5\text{O}_{12}$. ○: as-received state; ■ after annealing; thick broken line: Slack et al. [3]. Curve 1 is calculated assuming $\tau_N^{-1} = \tau_{Ex}^{-1} = 0$ (see text); curve 2 corresponds to $\tau_N^{-1} = 0$ and $\tau_{Ex}^{-1} \neq 0$; curve 3 corresponds to $\tau_N^{-1} \neq 0$ and $\tau_{Ex}^{-1} \neq 0$.

- iii) the entropy was independently measured.

We previously published a preliminary paper

on some thermal conductivity results [8]. A more complete analysis will be given here and calculations of the entropy and of the specific heat will be developed.

2. Sample characteristics and experimental results

The samples, provided by CRISMATEC, are monocrystals grown by the Czochralski method and have the eutectic composition which is slightly different from the stoichiometry. The general formula is assumed to be $Gd_3(Gd_xGa_{2-x})Ga_5O_{12}$ with x between 0.03 and 0.06, so that there is an excess of Gd compared with the stoichiometric composition. These excess Gd atoms are in octahedral sites, substituting for Ga atoms.

In the particular case of the samples which were used for specific heat and entropy measurements, an analysis by electron stimulated fluorescence was performed and x was found to be 0.058 ± 0.015 . Furthermore, an analysis by mass spectrometry was also done and the main impurities found are listed in table 1.

The sample used for specific heat and entropy measurements was cylindrically shaped ($\varnothing = 10$ mm, $l = 10$ mm). The crystallographic orientation was checked and it was found that there was a 4-fold axis oriented at $\theta = 34^\circ$ relative to the axis of the cylinder. For experiments at lower temperatures ($T < 1.2$ K), it was necessary to cut the sample in order to reduce its weight and to increase the area/volume ratio, since the specific heat was very high and it was consequently very difficult to cool the sample.

The samples used for the thermal conductivity measurements were the same as listed as I and II in ref. [8], parallelepipeds of dimensions $50 \times 4 \times 4$

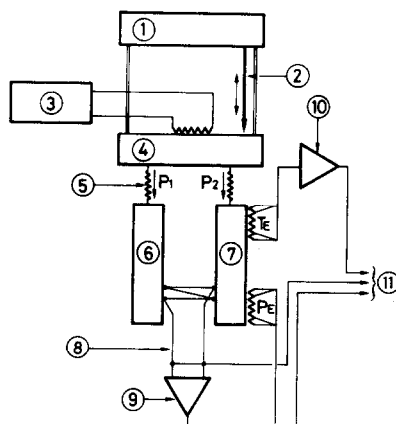


Fig. 2 Principle of operation of the apparatus for measurement of specific heat in the 0.3–20 K temperature range. 1 – Heat sink (0.3 K); 2 – mechanical thermal contact; 3 – microprocessor; 4 – controlled temperature plate; 5 – thermal resistance; 6 – reference cell, 7 – sample cell; 8 – differential thermocouple; 9 – compensation power amplifier; 10 – sample cell thermometer; 11 – to the data processing chain.

mm^3 and $20 \times 2.63 \times 2.57$ mm^3 , respectively.

All specific heat experiments were performed using a dynamic differential method [9]. In the particular case of the apparatus used for the measurements at the lowest temperatures, the heat sink is a pumped 3He bath coupled to a pumped 4He cooling stage [10].

The principle of the measurement is shown in fig. 2:

A plate (4) whose temperature is controlled by a microprocessor (3) is coupled to the heat sink (1) through a mechanical thermal contact (2) operated by a bellows thermally anchored at the 4He stage. Two identical measurement cells, (6) and (7), are coupled to the plate and receive a power P_1 and P_2 , respectively. The difference between P_1 and P_2

Table 1
Concentration of impurities given in ppm by weight

Element	Li	B	Na	Mg	P	S	Cl	K	Ca	Ti	V	Cr
Concentration	25	10	15	25	1	10	100	15	40	1.5	0.15	5
Element	Mn	Fe	Ni	Co	Cu	Zn	Rb	Y	Nb	Ba	Eu	
Concentration	1	75	1	≤ 0.7	10	13	0.4	2.5	≤ 1	1.3	≤ 2	

corresponds to the asymmetry of the thermal losses of the two cells and is assumed to be small. During the experiment, the plate is isolated from the heat sink and the power provided to the two cells is increased.

A temperature difference between the cell containing the sample and the reference cell is measured and additional power is supplied to the sample cell in order to compensate for this temperature difference.

The heat capacity of the sample is then given by:

$$C_s = \frac{P_s + (P_2 - P_1)}{v} + C_1 - C_2,$$

where P_s is the compensating electrical power and $C_1 - C_2$ the heat capacity difference of the two experimental cells when they do not have exactly the same temperature. $v = dT/dt$ is defined as the rate at which the temperature increases.

The results are shown in fig. 3. A peak is observed at $T = 0.9$ K but two slope changes are noticeable at $T = 0.7$ K and $T = 1.2$ K indicating clearly that there is an additional contribution in this temperature range. This point will be further

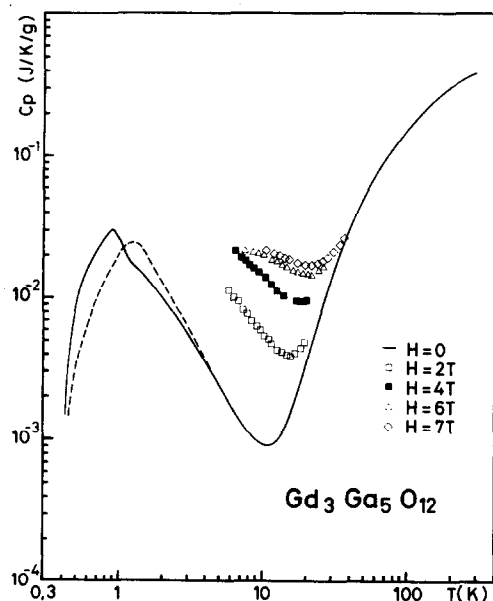


Fig. 3. Specific heat of $Gd_3Ga_5O_{12}$. The broken line is the result for $H = 0$ after annealing.

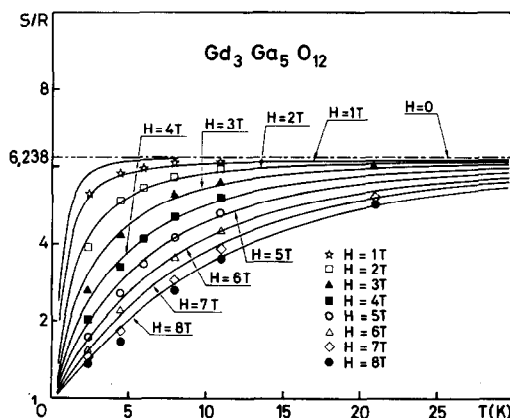


Fig. 4. Entropy of $Gd_3Ga_5O_{12}$. The full lines are calculated.

discussed below. The other observed feature is the large increase of C when a magnetic field is applied.

The entropy was measured independently of the specific heat by a method described elsewhere [11] and the results are shown in fig. 4 where S/R is plotted. The method enables us to measure an entropy difference. Clearly this means that the zero magnetic field entropy is not measurable and that the plotted values are dependent on the calculated value of this quantity.

In any case, the relative position of the measured entropy lines with respect to each other is independent of the reference and provides the information of interest.

The accuracy of the measurements was checked and was found to be better than 5% in all temperature ranges.

The thermal conductivity $K(T)$ has been measured from 60 mK to 20 K and with an applied magnetic field up to 7 T using the stationary flux method, which gives a systematic error in $K(T)$ of less than 7%. Most of the results have already been published [8]. In fig. 1 the zero field conductivity $K(T)$ of sample II in the as-received state is reported. Results are compared with previous measurements carried out by Slack et al. [3]. It can be seen that the temperatures T_m of the maxima are almost identical (≈ 12 K) as are the magnitudes of $K(T)$ at T_m . In contrast, below T_m the thermal conductivity measured by Slack was lower by a

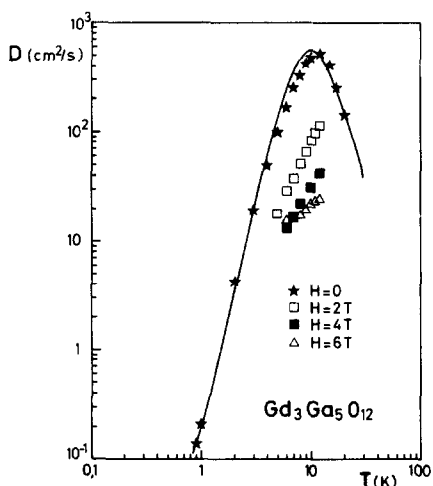


Fig. 5. Diffusivity of $\text{Gd}_3\text{Ga}_5\text{O}_{12}$. The full line is calculated.

factor of ≈ 5 than that found in our measurements.

The thermal diffusivity coefficient, D , was deduced from specific heat and thermal conductivity measurements.

It is defined as:

$$D = K/C\rho,$$

where K is the thermal conductivity, C the specific heat and ρ the density of the material. The variation of D as a function of the temperature is plotted in fig. 5.

D has the dimensions of a surface divided by a time and is easily related to a thermal time constant measuring the speed with which heat may propagate in the system.

At $T = 1.8$ K, the zero magnetic field diffusivity is equal to 2.4. For a characteristic length of 1 cm, this corresponds to a 0.42 s time constant.

As observed in fig. 5, D varies roughly as T^4 from 1 to 5 K and goes through a maximum at $T \approx 12$ K. A consequence of such a temperature dependence is that time constants are expected to be very short at high temperature.

Nevertheless, D is considerably reduced by application of a magnetic field. The result for $H = 7$ T is plotted in fig. 5. The results for $H = 2$ and 4 T are also plotted assuming that $K(T)$ did not vary with magnetic field. Strictly speaking this is not correct but since C has a larger field dependence

than K , it is possible to obtain in this way a rough estimation of D .

3. Theory

The ground state of the Gd^{3+} ion is an $^8S_{7/2}$ state which is eight-fold degenerate.

This degeneracy is removed by the electric crystalline field of cubic symmetry and the result is a Γ_7 ground state doublet, a Γ_8 quadruplet and a Γ_6 doublet. A magnetic field will remove the remaining degeneracies and the resulting splittings have been calculated by various authors for particular orientations [12,13].

In the general case, when the magnetic field has an arbitrary direction with respect to the crystallographic axis of the system (see inset of fig. 6), the energy levels are obtained from the diagonalization of a secular equation given by Lacroix [13]. The only adjustable quantities in the calculation are the crystal field parameters δ and ϵ (in the notation of Lacroix). g was taken to be equal to 2 and was considered to be isotropic.

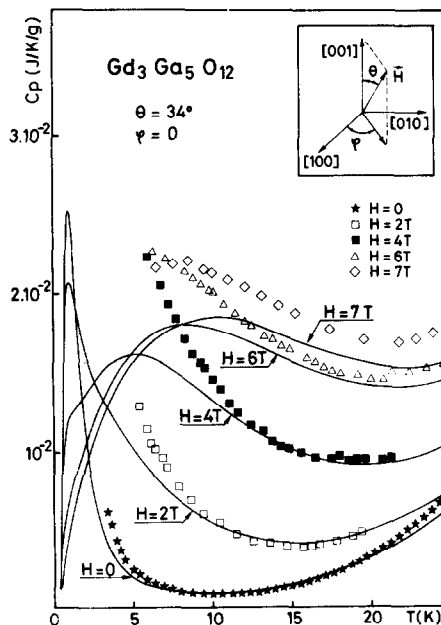


Fig. 6. Fit of specific heat for $H = 0$ and for various magnetic fields. Inset: Orientation of the magnetic field with respect to the crystallographic axes.

In this approximation, the inequivalencies of the sites of the rare earth ions, leading to inequivalent orientations of the g tensor for each site, are neglected [14]. The g value has been measured along various crystallographic directions [5–7] and found to be very close to the $g = 2.0023$ value corresponding to the theoretical value for a free spin [15]. Therefore the weakness of the observed anisotropy justifies the approximation.

The diagonalization of the secular equation was computed numerically and the result is a set of energy values, $W_i(H, \theta, \phi)$, i running from 1 to 8.

The entropy of the system is given by:

$$S(T, H, \theta, \phi) = k \left[N \log N - \sum_i N_i \log N_i \right], \quad (1)$$

where $N = \sum_i N_i$ = total number of Gd^{3+} ions and $N_i = N \exp[-W_i(H, \theta, \phi)/kT]/Z$.

In this last expression, Z is the partition function of the system and is given by $\sum_i \exp[-W_i/kT]$ where k is the Boltzmann constant.

The specific heat is then deduced from the entropy:

$$C(T, H, \theta, \phi) = T dS/dT. \quad (2)$$

The important feature of this calculation is that both the entropy and the specific heat depend upon the magnetic field orientation. A convenient choice of geometry must then be carefully examined in order to optimize a magnetic refrigerator working with this compound since the effect was found to be quite strong in the temperature range 0–5 K.

The position of the maximum of specific heat at low temperature is very sensitive to the value of δ , which enabled us to determine this parameter. This fit was found not to be very sensitive to the value of ϵ .

The lattice term was fitted once δ and ϵ had been properly determined.

The result is shown in fig. 6. The best parameters are $\delta = 0.43$ K, $\epsilon = 0.13$. The lattice term was found to be $C_L = 3.95 \times 10^{-7}$ J/gK.

Although the position of the specific heat maximum is correctly given by the calculation, the agreement between experiment and theory is poor so far as the value of C is concerned at this

temperature. This is not surprising since there is experimental evidence that an additional contribution to C is present in this temperature range. If C is smoothly extrapolated between 0.7 and 1.2 K it is possible to estimate how large this extra specific heat is compared to the total anomaly. It was found that it was 4% of the total specific heat anomaly, that is of the same order of magnitude as the number of Gd^{3+} ions in excess in the material.

In order to test this hypothesis, the sample was annealed for 12 h at 1650°C in the presence of oxygen (the fusion temperature of GGG is 1750°C) and the specific heat was subsequently measured. The result is shown in fig. 3, and two remarkable features are noticeable:

- i) the extra contribution disappeared and the value of the specific heat at the maximum was reduced,
- ii) the temperature of the maximum was shifted to 1.2 K.

These results are evidence that there are actually oxygen vacancies in the material. The total enthalpy integrated over all the specific heat anomaly was found to be 6% smaller after annealing than in the as-received state. Although the value of the specific heat is then in good agreement with the theory, it was not possible to fit the experimental curve by changing δ or ϵ .

When a magnetic field is applied, the specific heat calculated with eq. (1) and (2) is found to be much larger than the experimentally measured value. Such a discrepancy is also observed for the entropy. The reason is that the interactions between the Gd^{3+} ions were not taken into account in the calculation.

The susceptibility of $Gd_3Ga_5O_{12}$ was measured by Wolf et al. [16] and was found to obey a Curie–Weiss law at high temperature:

$$\chi = C_c / (T - T_c),$$

with the molar Curie constant $C_c = 7.82$ and $T_c = -2.3$ K.

These measurements indicate that there is a strong antiferromagnetic coupling between the Gd^{3+} ions. The variation of χ was confirmed by Cooke et al. [17] and in both experiments a departure from the Curie–Weiss law was observed at

low temperature (below 5 K in the Wolf et al. experiment).

In a molecular field approach, corresponding to the Curie-Weiss approximation, it is possible to calculate the magnetic dipole interaction field as a function of the applied magnetic field, H , in the paramagnetic phase.

It is found that $H_e = [T_c/(T - T_c)] H$, so that the total field is

$$H_a = H + H_e = H[1 + T_c/(T - T_c)]. \quad (3)$$

The calculation of the entropy and of the specific heat by means of eqs. (1) and (2) using the above expression for H_a is shown in figs. 4 and 6.

The fit of the specific heat is reasonably good for temperatures above 10 K. The disagreement at low temperatures is obviously due to the failure of the molecular field approximation when short-range order interactions become important. A microscopic analysis by Kinney et al. [18] investigated the role played by interactions with first, second and third nearest neighbors. J_1 and J_3 were found to be of antiferromagnetic type whereas J_2 is of ferromagnetic type.

The entropy calculated with eqs. (1), (2) and (3) is a good fit to the experimental results as shown in fig. 4 but the same discrepancy as for the specific heat was observed at low temperature.

It is worth noting that these interactions should be taken into account in the interpretation of the zero magnetic field specific heat data, and that the adjustment of the crystal field parameters which leads to a good agreement of these data with the theory could include and mask these effects.

Calculations of the thermal conductivity have been performed using the full Callaway model [19]:

$$K(T) = K_1(T) + K_2(T),$$

where $K_2(T)$ is a correction taking into account the peculiar nature of the N three-phonon processes.

$K_1(T)$ is the classical Debye expression:

$$K_1(T) = \frac{k}{2\pi^2 v} \left(\frac{kT}{\hbar} \right)^3 \int_0^{\theta_D/T} \frac{x^4 e^x}{(e^x - 1)^2} \tau(\omega, T) dx,$$

where $x = \hbar\omega/kT$, θ_D is the Debye temperature

(520 K) and v the mean sound velocity (3.97×10^3 m/s). $\tau(\omega, T)$ is the combined relaxation time obtained by adding the reciprocal relaxation times associated with the appropriate scattering mechanisms:

$$\tau^{-1}(\omega, T) = A\omega^4 + G\omega + \frac{v}{\alpha L} + \tau_u^{-1} + \tau_N^{-1} + \tau_{ex}^{-1},$$

τ_{ex}^{-1} describes the scattering due to extrinsic defects. The other terms are generally present in any insulating crystal.

τ_u^{-1} and τ_N^{-1} account for the 3-phonon processes. They depend on both T and ω and are efficient around and above T_m .

The three remaining terms describe non-resonant scattering by point defects ($A\omega^4$), dislocations ($G\omega$) and boundaries ($v/\alpha L$), and limit the mean free path of phonons at $T \lesssim T_m$, the temperature range of particular interest in this study.

$v/\alpha L$ can be theoretically calculated from the crystal size and the mean sound velocity α is a coefficient taking into account the nature of phonon reflections [20] (assumed to be perfectly diffuse, since the surface is rough). We obtain:

$$v/\alpha L = 1.35 \times 10^{-6} \text{ s}^{-1}.$$

GGG is a compound which can be grown free of dislocations ($N_d < 1/\text{cm}^2$), thus we are able to set $G = 0$.

It is possible to calculate A , provided that the chemical composition of the sample is known [21]. According to the mass spectrometer analysis, about 300 ppm of different impurity atoms are present. This is negligible in comparison with the mass defect due to non-stoichiometry. Using the value $x = 0.058$, A was found to be $7.5 \times 10^{-44} \text{ s}^{-3}$.

Using these parameters, the theoretical curve calculated setting $\tau_u^{-1} = \tau_N^{-1} = \tau_{ex}^{-1} = 0$, is shown in fig. 1.

The two main remarks are as follows:

- At very low temperatures ($T < 0.1$ K) the measured thermal conductivity reaches the calculated limit. This point rules out Slack's suggestion that only longitudinal phonons could contribute to the heat transport.
- Around T_m it is impossible to fit the data using the calculated point defect scattering term only.

Both magnitude and shape display discrepancies.

In order to account for the dip in $K(T)$ above $T = 0.1$ K, a resonant Lorentzian relaxation time was introduced:

$$\tau_{\text{ex}}^{-1}(\omega, T) = D\omega^2 \frac{\Gamma}{\Gamma^2 + (\omega - \omega_E)^2},$$

with $\omega_E = 0.75$ K, $D = 4 \times 10^6 \text{ s}^{-1}$ and $\Gamma = 0.25$ K. The results are plotted as curve 3 in fig. 1.

The good agreement, up to 4 K, between data and calculations clearly indicates that only one defect is responsible for this resonant scattering. In order to attempt to identify the defect and to check if this effect is correlated with the anomalous specific heat "bump" observed in the as-received state, the sample was annealed in the same conditions as previously described. Below 1 K, in contrast with the change observed in specific heat, no detectable effect was found in $K(T)$ (fig. 1). Thus, this is an indication that this extra scattering is not related to the deviation from stoichiometry. Furthermore, the relatively small change in $K(T)$ observed at $T > 1.2$ K for $H = 7$ T is another indication that Gd^{3+} ions are not the ions mainly responsible for this extra scattering, as the energy levels of the system are widely split by the magnetic field.

Unfortunately, it is not possible to identify clearly at this stage of the study, the defect to which the extra scattering could be attributed.

Above 4 K, the disagreement between experiment and calculation indicates that additional scattering processes are present. N -processes are the most likely but it is not possible to discount a scattering mechanism due to macroscopic inclusions or to the energy levels of impurities.

It is possible to achieve a good fit between 4 and 20 K assuming tentatively that only N processes are present and that the associated inverse relaxation time is:

$$\tau_N^{-1} = 7.6 \times 10^{-9} \omega T^2 \text{ s}^{-1},$$

but such a fit is quite phenomenological.

The theoretical expression for the diffusivity was simply deduced from the previously calculated specific heat and thermal conductivity (including

τ_N^{-1} in the total inverse relaxation time).

The best fit is shown in fig. 5. The agreement is particularly good from 0.8 to 2 K and the discrepancy between experiment and the calculated value never exceeds 50% in the whole range of temperatures.

4. Conclusion

In conclusion, it appears that the crystals we used were of a far better quality than samples which had previously been studied. The various properties we measured have been coherently bound together and qualitatively understood.

In addition, a quantitative model has been developed and provides satisfactory agreement with the experiments for T higher than 5 K and for all values of magnetic field.

Below 5 K, the discrepancies are clearly a result of the simplicity of the model but the experimental data are sufficiently accurate in this temperature range to provide a useful tool to calculate the parameters of interest in magnetic cooling.

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References

- [1] C. Delpuech, R. Beranger, G. Bon Mardion, G. Claudet, and A.A. Lacaze, *Cryogenics* 21 (1981) 579.
- [2] A.F. Lacaze, R. Beranger, G. Bon Mardion, G. Claudet, C. Delpuech, A.A. Lacaze and J. Verdier, *Proc. CEC* 81, San Diego (1981).
- [3] G.A. Slack and D.W. Oliver, *Phys. Rev. B* 4 (1971) 592.
- [4] D.G. Onn, H. Meyer and J.P. Remeika, *Phys. Rev.* 156 (1967) 663.
- [5] R.A., Fisher, G.E. Brodale, E.W. Hornung and W.F. Giauque, *J. Chem. Phys.* 59 (1973) 4652.

- [6] E.W. Hornung, R.A. Fisher, G.E. Brodale and W.F. Giauque, *J. Chem. Phys.* 61 (1974) 282.
- [7] G.E. Brodale, W.F. Hornung, and W.F. Giauque, *J. Chem. Phys.* 62 (1975) 4041.
- [8] B. Daudin and B. Salce, *Compt. Rend. Acad. Sci. (France)* 293 (1981) 885.
- [9] R. Lagnier, J. Pierre and M.J. Mortimer, *Cryogenics* 17 (1977) 341.
- [10] E. Bonjour, R. Lagnier, M. Locatelli and J. Pierre, *Proc. Journées de Calorimétrie et d'Analyse Thermique, Barcelone* (1980).
- [11] B. Daudin, R. Lagnier and B. Salce, *J. Magn. Magn. Mat.* 25 (1981) 197.
- [12] J. de Boer and R. van Lieshout, *Physica* 15 (1949) 569.
- [13] R. Lacroix, *Helv. Phys. Acta* 30 (1957) 374.
- [14] W. Low and E.L. Offenbacher, *Solid State Phys.* 17 (1965) 135.
- [15] A. Abragam and B. Bleaney, *Résonance paramagnétique électronique des ions de transition* (PUF, Paris, 1971) p. 436.
- [16] W.P. Wolf, M. Ball, M.T. Hutchings, M.J.M. Leask and A.F.G. Wyatt, *J. Phys. Soc. Japan* 17 Suppl. B.I. (1962) 443.
- [17] A.H. Cooke, T.L. Thorp and M.R. Wells, *Proc. Phys. Soc.* 92 (1967) 400.
- [18] W.I. Kinney and W.P. Wolf, *Proc. 24th Ann. Conf. on Magn. and Magn. Mat. Munich* (1979).
- [19] J. Callaway, *Phys. Rev.* 113 (1959) 1046.
- [20] R. Berman, F.E. Simon and J.M. Ziman, *Proc. Roy. Soc.* 220 (1953) 171.
- [21] P.G. Klemens, *Solid State Phys.* 7 (1958) 1.